What is a parameter Variable :
----
* If we declare a variable as a **method parameter** (not inside the method body) then It is called parameter variable

Example :

```
public void deposit(double amount)
{
    if(amount <=0)
    {
     IO.println("Invalid Amount");   //Validation of data
     return;
    }
```

* It is used to receive the value from the **outer world** (End Client).

* Generally, We perform validation for parameter variable.

* Just like local variable, parameter variables also accessible within the same method body only.

WAP to find out the product of a given number :
Example :
```
          32 = 6
          54 = 20
         167 = 42

 void main()
 {
      int num = Integer.parseInt(IO.readln("Enter a number :"));
      IO.println("Product of the "+num+" is :"+getProductOftheNumber(num));
 }

 int getProductOftheNumber(int num)  //num = 0  [1 * 2 * 3]
 {
      int product = 1;  //6

    while(num !=0)
      {
           int digit = num % 10;  //digit = 1
           product = product * digit;
           num = num /10;
      }

    return product;

 }
```

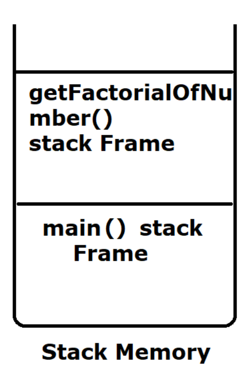
4) WAP to find out the largest digit from the given number.
   Example :
```
             189  : 9
              54  : 5
             842  : 8

 void main()
 {
      int number = Integer.parseInt(IO.readln("Enter a number :"));
      IO.println("Largest digit in "+number+" is = "+getLargestDigit(number));

 }

 int getLargestDigit(int num)
 {
   int max = 0; //9

   while(num !=0) // 0
      {
        int digit = num % 10;  //digit = 9

      if(digit > max)   //9 > 7
          {
               max = digit;
          }
       num = num /10;
      }

    return max;
 }
```

5) WAP to find out the smallest digit in a given number.
   Example :
```
            54  :  4
          123   :   1

    void main()
    {
         int number = Integer.parseInt(IO.readln("Enter a number :"));
         IO.println("Smallest digit in "+number+" is = "+getSmallestDigit(number));

    }
    int getSmallestDigit(int num)  //123
    {
         int min = 9; //1

       while(num !=0) //num = 0
         {
              int digit = num % 10;  //digit = 1

              if(digit < min)  //1 < 2
              {
                   min = digit;
              }
          num = num /10;
         }

       return min;

    }
```

6) Find the factorial of a given number :
    Example : !5 = 120 [5 X 4 X 3 X 2 X 1 = 120]
```
void main()
{
     int number = Integer.parseInt(IO.readln("Enter a Number :"));
     IO.println("Factorial of "+number+ " is :"+getFactorialOfNumber(number));
}

int getFactorialOfNumber(int num)
{
  int fact = 1;

  for(int i=1; i<=num; i++)
     {
       fact = fact * i;
     }

     return fact;
}
```

Recursive Function OR Method :
----
*A recursive function is a function that **calls itself** either directly or indirectly and must contain a **base condition** to stop execution.

* In recursive method, id we don't provide any based condition then It will keep on running as a result
  Stack memory is not sufficient to hold more methods hence java.lang.StackOverflowException will be generated



Stack Memory

```
 void main()
 {
      int number = Integer.parseInt(IO.readln("Enter a Number :"));
      IO.println("Factorial of "+number+ " is :"+getFactorialOfNumber(number));
 }

 int getFactorialOfNumber(int num) //num = 4
 {
      if(num ==1)   //base condition
      {
           return 1;
      }

   return num * getFactorialOfNumber(num -1);
 }
```

8) WAP to find out the reverse of a number in **String format.**
```
  void main()
  {
       int number = Integer.parseInt(IO.readln("Enter a Number :"));
       IO.println("Reverse of "+number+" is :"+getReverseOfANumber(number));

  }

  String getReverseOfANumber(int num)  //789
  {
    String reverse = ""; //""987

    while(num!=0) //num = 0
    {
      int digit = num % 10; //digit = 7
      reverse = reverse + digit;
        num = num /10;
    }
    return reverse;


  }
```

9) WAP to reverse a number in number format.

    Example :
```
                 567  = 765
                 123  = 321

 void main()
 {
      int number = Integer.parseInt(IO.readln("Enter a Number :"));
      IO.println("Reverse of "+number+" is :"+getReverseOfANumber(number));

 }

 int getReverseOfANumber(int num)  //123            -> 321
 {
   int reverse = 0;

   while(num !=0)
   {
        int digit = num % 10;
        reverse = (reverse * 10) + digit;
        num = num /10;
   }

    return reverse;
 }

 /*
 num = 123

 digit =  3   rev = 3            //(rev * 10) + digir //3
 digit =  2   rev = 32           //(rev * 10) + digit  //32
 digit =  1   rev = 321          //(rev * 10) + digit  //321

 */
```